

Refraction of Waves

Why a wave changes speed, wavelength and direction on crossing into a new medium — while its frequency never changes.

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PHYSICS

Grade 11 · Egyptian Baccalaureate

10 questions

Step-by-step solutions

f is the same in both media

The toolkit for this lesson

$$v = f \lambda$$

and f is set by the **source**

$$f_1 = f_2$$

the frequency never changes on refraction

$$v_1/v_2 = \lambda_1/\lambda_2$$

so speed and wavelength change together

$$\sin i / \sin r = v_1 / v_2$$

the law of refraction

$$n = c / v$$

a slower medium has a larger n

shallow water

slower waves, shorter λ , same f

One thing is fixed and everything else follows. The **frequency** of a refracted wave is fixed by the source, so it is the same in both media. Since $v = f\lambda$, the speed and the wavelength must therefore change **by the same factor** — and it is that change of speed that bends the direction of travel.

Sea waves approaching the shore

EXPLAIN

- 1 The figure shows the change in the shape, the direction, the behaviour and the properties of sea waves as they travel from deep water into shallow water near the shore. Using the figure and what you have studied about the properties of waves:

- Explain the changes that take place in each of: the speed of the waves, their wavelength, and their direction of propagation, as they pass from deep water into shallow water.
- Explain the reason for the bending of the waves and the adjustment of their direction so that they become parallel to the shore line near the beach.
- Deduce the physical concept that this phenomenon describes, and state its definition in your own words.

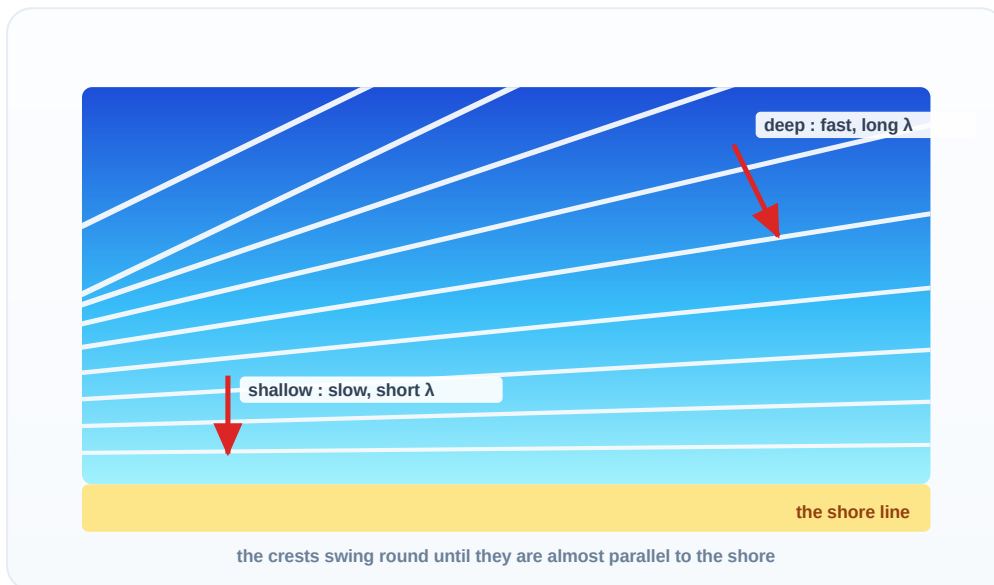


FIGURE 1

- 2 Which of the following changes take place in sea waves as they pass from deep water into shallow water near the shore?

MCQ

- A** their speed increases, their wavelength increases, and they continue in a straight line without any bending
- C** their speed stays constant, their wavelength decreases, and they bend away completely from the shore line

- B** their speed decreases, their wavelength decreases, and their direction bends so that they become parallel to the shore line
- D** their speed decreases, their wavelength increases, and they are completely reflected back towards the deep water

Refraction in general

EXPLAIN

- 3 The figure shows the refraction of sea waves at different coastal shapes. Given that the depth of the water at the headland is less than its depth at the bay, **draw a table comparing the properties of the water waves at the “headland” and at the “bay”** as regards:

- a) the wave speed (v).
- b) the wavelength (λ).
- c) the wave frequency (f).

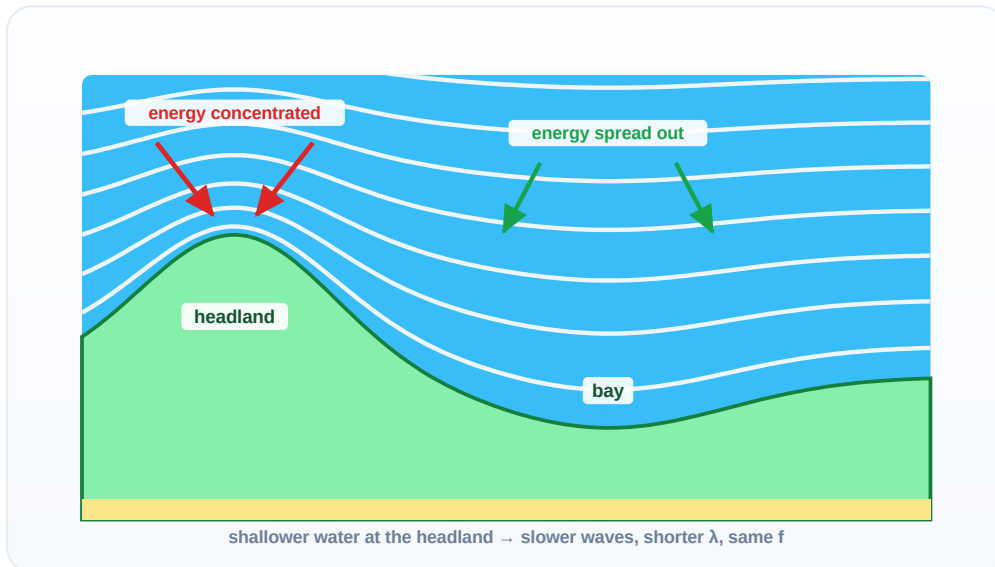


FIGURE 2

- 4 Which of the following statements correctly describes what happens to a light wave or a water wave when it is refracted while passing at an oblique angle from one medium into another medium of different properties?

MCQ

A the frequency of the wave changes, while its speed and its wavelength stay constant

B the speed of the wave and its wavelength change, while its frequency stays constant

C the speed of the wave, its wavelength and its frequency all increase

D the speed of the wave, its wavelength and its frequency all decrease

- 5 The graph shows the relation between the speed of a wave (v) and its wavelength (λ) for three waves from different sources, as the wave passes between media of different properties. **Which of the frequencies has the largest value?** Give the explanation.

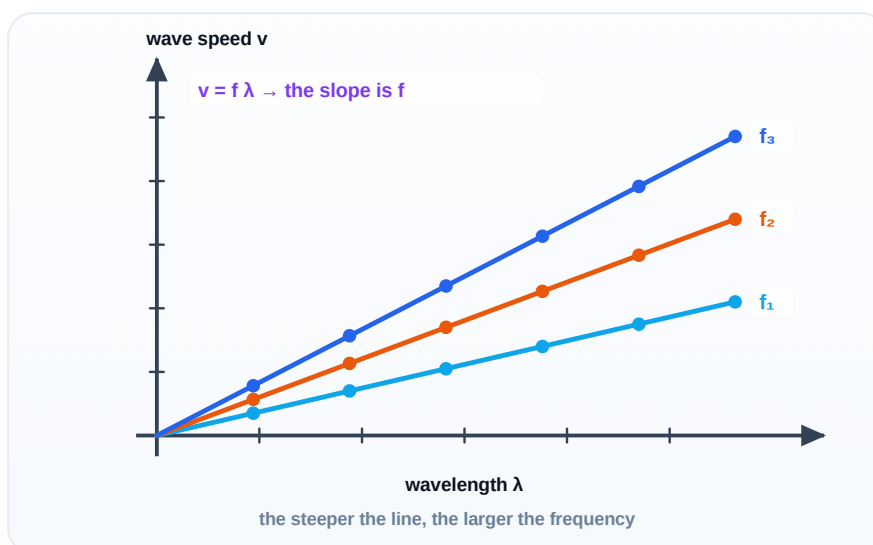


FIGURE 3

- 6 When a wave spreads through vacuum or through a medium, the locus of all the points that vibrate in the same phase is called the

MCQ

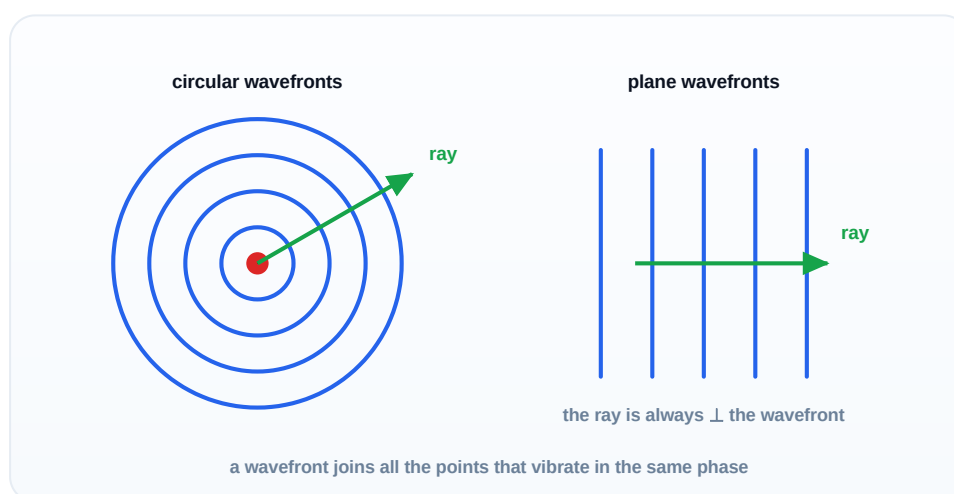


FIGURE 4

A frequency of the wave

B wavelength

C speed of the wave

D wavefront

- 7 A wave crosses a boundary separating two media as shown in the figure. **Calculate the following:**

- a) the ratio (the frequency of the wave in medium 1) / (the frequency of the wave in medium 2) =
- b) the ratio (the wavelength of the wave in medium 1) / (the wavelength of the wave in medium 2) =

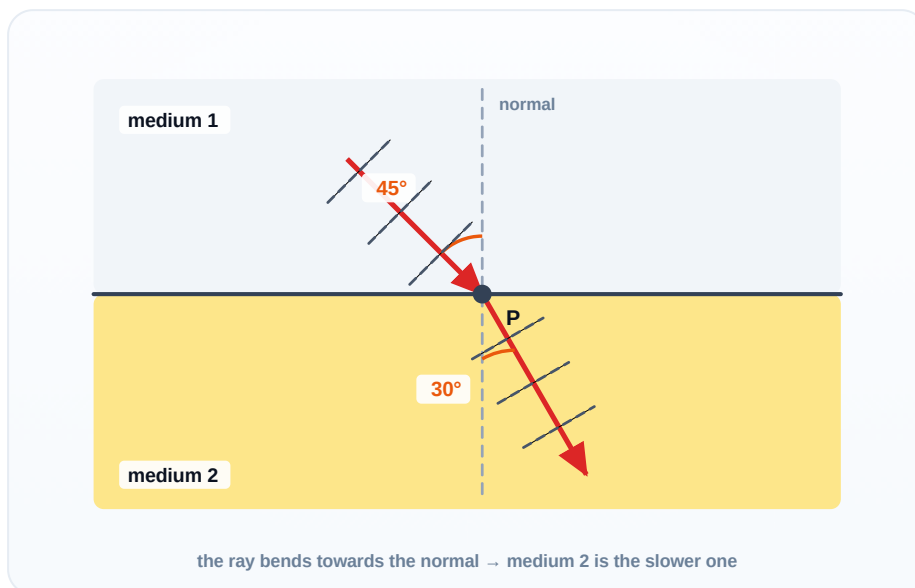


FIGURE 5

- 8 A wave of frequency 50 Hz travels in medium (1) with a speed of 300 m/s. On passing into medium (2) its speed changes to 600 m/s. Which of the following statements is **not** correct?

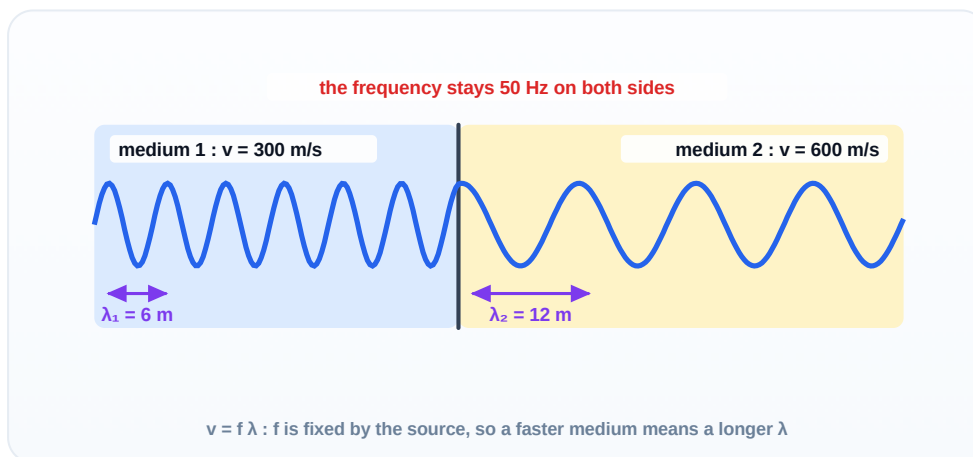


FIGURE 6

A the frequency of the wave in medium (2) equals 50 Hz

B the wavelength of the wave in medium (1) equals half its value in medium (2)

C the angle of incidence of the wave in medium (1) equals half the angle of refraction in medium (2)

D the refractive index of medium (1) equals twice the refractive index of medium (2)

MCQ

9 A light ray passes at an oblique angle from medium (1) into medium (2). Given that the absolute refractive index of medium (1) is greater than the absolute refractive index of medium (2), that is $n_1 > n_2$, which of the following physical changes correctly describes what happens to the light ray on entering medium (2)?

A the ray is refracted towards the normal, and its speed and wavelength decrease

B the ray is refracted away from the normal, and its speed and wavelength increase

C the ray is refracted away from the normal, its speed increases and its frequency decreases

D the ray passes straight on without any deviation, and its speed and wavelength stay constant

MCQ

10 A light ray travels from material (A) into material (B), so that the angle of incidence in material (A) equals 30° and the angle of refraction in material (B) equals 45° . Which of the following statements correctly describes the behaviour of the light and the physical properties of the two materials?

A the refractive index of (A) is greater than that of (B), and the speed of light in (A) is less than it is in (B)

B the refractive index of (A) is less than that of (B), and the speed of light in (A) is greater than it is in (B)

C the refractive index of (A) is greater than that of (B), and the speed of light in (A) is greater than it is in (B)

D the refractive index of (A) equals that of (B), and the frequency in (A) is doubled in (B)



Solutions — step by step

Start every refraction question the same way. Write $f_1 = f_2$. Then $v = f\lambda$ forces λ to follow the speed: a slower medium means a shorter wavelength, and the ray bends **towards** the normal.

Part A — Classwork 1

1 sea waves running into shallow water

a) **The three changes.** The speed of a water wave depends on the **depth**: the shallower the water the slower the wave, so on moving inshore **v decreases**. The frequency is set by whatever produced the waves far out at sea, so **f does not change**. Since $v = f\lambda$ and f is fixed, the wavelength must fall in the same proportion as the speed: **λ decreases**, which is why the crests are drawn closer and closer together in the figure. The **direction** swings round until the crests are almost parallel to the shore line.

b) **Why they bend.** A crest arriving obliquely does not meet the shallow water all at once. The end of the crest that is already in shallow water slows down, while the other end is still in deeper water and keeps moving faster. The faster end therefore swings forward about the slower one, exactly as a marching line wheels about its inner end, and the crest is turned until it lies along the depth contours — that is, along the shore line.

c) **The concept.** This is **refraction**: the change in the direction of a wave when it passes obliquely from one medium into another in which its speed is different, its frequency staying the same.

a) v decreases, λ decreases, f stays the same, and the direction bends towards the shore b) because the shallow end of each crest slows down first c) refraction

2 what changes from deep to shallow water

depth decreases $\rightarrow v$ decreases

f is fixed by the source, so from $v = f\lambda$:

$$\lambda = v / f \quad \rightarrow \lambda \text{ decreases as well}$$

and the oblique crests swing round towards the shore line.

Choice (D) is the tempting wrong one: it gets the speed right but then makes λ increase, which would need the frequency to fall — and the frequency cannot change.

B) the speed decreases, the wavelength decreases, and the direction bends to become parallel to the shore line

Part B — Home assignment 1

3 the headland compared with the bay

the water at the headland is **shallower**, so the waves there are slower :

$$v(\text{headland}) < v(\text{bay})$$

the frequency is the same everywhere (the same source out at sea) :

$$f(\text{headland}) = f(\text{bay})$$

so from $\lambda = v / f$:

$$\lambda(\text{headland}) < \lambda(\text{bay})$$

Look at the figure: because the crests slow down over the shallow water off the headland, they bend round it from both sides. The rays therefore **converge** on the headland and **diverge** in the bay. That is why headlands are attacked and worn away by the sea while bays stay calm and collect sand — the wave energy per metre of coast is far greater at the headland.

property	at the headland (shallow)	at the bay (deeper)
wave speed v	smaller	larger
wavelength λ	shorter	longer
frequency f	the same	the same
wave energy	concentrated	spread out

the headland has the smaller v and the shorter λ , and the same f as the bay

4 what refraction changes and what it does not

the source sets the frequency, and the boundary cannot create or destroy cycles :

$$f_1 = f_2 \quad (\text{the frequency never changes})$$

the new medium sets a new speed, so from $v = f \lambda$ with f fixed :

$$v \text{ changes} \rightarrow \lambda \text{ changes by the same factor}$$

Think of the boundary as a turnstile: the same number of crests arrive every second as leave every second, otherwise crests would pile up at the boundary. That is the whole reason why f cannot change.

B) the speed of the wave and its wavelength change, while its frequency stays constant

5 which frequency is the largest

the wave equation $v = f \lambda$ can be read as the equation of a straight line :

$$v = f \times \lambda \rightarrow (y = \text{slope} \times x)$$

so for each source the graph of v against λ is a straight line through the origin whose **slope is the frequency**.

the steepest of the three lines is the one belonging to f_3 .

Read one point off any line to get a number: if a line passes through $\lambda = 4.0$ m at $v = 200$ m/s, then $f = v/\lambda = 50$ Hz. The three lines all pass through the origin, which is the graphical way of saying that a wave of zero wavelength would have zero speed.

f_3 has the largest value, because its line has the greatest slope and the slope of a $v - \lambda$ graph is the frequency

6 the name of the locus of points in the same phase

all the points of a **wavefront** have been disturbed by the source for the same length of time, so they are all at the same stage of their vibration. near a point source the wavefronts are circles (or spheres) ; far away they are practically straight lines (or planes). the ray is always $\perp$ the wavefront.

D) the wavefront

7 the two ratios at the boundary

from the figure : $i = 45^\circ$ in medium 1 $r = 30^\circ$ in medium 2

a) the frequency is set by the source and cannot change at a boundary :

$$f_1 / f_2 = 1$$

b) since f is the same, $\lambda \propto v$, and the law of refraction gives :

$$\lambda_1 / \lambda_2 = v_1 / v_2 = \sin i / \sin r$$

$$\lambda_1 / \lambda_2 = \sin 45^\circ / \sin 30^\circ = 0.7071 / 0.5$$

$$\lambda_1 / \lambda_2 = 1.41 \quad (= \sqrt{2})$$

The ray bends **towards** the normal ($45^\circ \rightarrow 30^\circ$), so medium 2 is the slower, optically denser one — and a slower medium always carries the shorter wavelength, which is exactly what the ratio 1.41 says.

a) $f_1 : f_2 = 1 : 1$ b) $\lambda_1 : \lambda_2 = \sqrt{2} : 1 \approx 1.41$

8 which statement is not correct

check each statement against $f_1 = f_2 = 50 \text{ Hz}$ and $\lambda = v / f$:

$$\lambda_1 = 300 / 50 = 6 \text{ m}$$

$$\lambda_2 = 600 / 50 = 12 \text{ m}$$

(A) f in medium 2 = 50 Hz ✓ correct

(B) $\lambda_1 = 6 \text{ m} = \text{half of } \lambda_2 = 12 \text{ m}$ ✓ correct

(D) $n = c / v$, so $n_1 / n_2 = v_2 / v_1 = 600/300 = 2$ ✓ correct

(C) the law of refraction gives

$$\sin i / \sin r = v_1 / v_2 = 1 / 2, \text{ that is } \sin r = 2 \sin i$$

which is **not** the same as $i = r / 2$ ✗ wrong

The relation between the two angles involves their **sines**, not the angles themselves. Test it with numbers: for $i = 20^\circ$, $\sin r = 2 \sin 20^\circ = 0.684$, so $r = 43.2^\circ$ — not 40° .

C) — the statement that the angle of incidence equals half the angle of refraction

9 entering a medium of smaller refractive index

$$n_1 > n_2 \quad \text{and} \quad n = c / v \quad \rightarrow \quad v_1 < v_2$$

so the ray speeds up on entering medium 2, which means it bends **away** from the normal. with f unchanged, $\lambda = v / f$ increases as well.

Choice (C) says the frequency decreases — the one thing refraction can never do. Remember the pair: **into a denser medium** means slower, shorter λ , bending towards the normal; **into a less dense medium** means faster, longer λ , bending away from it.

B) the ray is refracted away from the normal, and its speed and wavelength increase

10 comparing the two materials A and B

apply the law of refraction between the two materials :

$$n(A) \sin i = n(B) \sin r$$

$$n(A) \sin 30^\circ = n(B) \sin 45^\circ$$

$$n(A) \times 0.5 = n(B) \times 0.7071$$

$$n(A) / n(B) = 0.7071 / 0.5 = 1.41 \quad \rightarrow \quad n(A) > n(B)$$

and since $v = c / n$:

$$\text{a larger } n \text{ means a smaller } v \quad \rightarrow \quad v(A) < v(B)$$

A quick sanity check without any algebra: the ray bends **away** from the normal on going from A into B ($30^\circ \rightarrow 45^\circ$), so it must have **sped up** — B is the faster, less dense material. Everything else follows from that one observation.

A) the refractive index of (A) is greater than that of (B), and the speed of light in (A) is less than in (B)